

Week 2: My First Automated Financial Analysis

- **Key Concepts:**

- **Algorithmic Financial Analysis:** Translating a mathematical formula (Altman Z-Score) into a reusable Python function.
- **Reactive Programming:** Introduction to **Marimo**, where changing an input (Ticker) automatically re-runs dependent code cells (unlike Jupyter's linear execution).
- **API Data Retrieval:** Moving away from manual data entry by programmatically fetching live financial statements (Balance Sheet, Income Statement) from the web.
- **Data Visualization:** Representing financial health visually (bar charts with threshold rules) rather than just numerically.
- **Panel Data Structure:** Moving beyond simple lists to structured data that covers multiple entities (companies) over multiple time periods (years)—a fundamental concept in financial research.
- **Robustness in Automation:** Understanding the need for scripts to handle missing data points without crashing (Exception Handling) and the importance of pacing requests to avoid being blocked (Rate Limiting).

- **Coding Techniques:**

- **Exception Handling:** Using `try / except` blocks to write code that skips errors gracefully in case of undefined values or missing data
- **Function Definition:** Creating a custom function (`def Zscore...`) to modularize logic.
- **Object Attribute Access:** Accessing specific financial lines using dictionary-like methods (`.get`) and pandas method for integer-location based indexing (`.iloc`).
- **User Inputs:** Binding a Python variable to a frontend input box (`mo.ui.text`).
- **Conditional Logic in Visualization:** mapping colors (Green/Orange/Red) dynamically based on calculated score thresholds.
- **Nested Loops:** Implementing a "Loop within a Loop" structure (Iterating through Years → Iterating through Tickers) to build a dataset systematically.
- **Rate Limiting:** Implementing `time.sleep()` to pause execution between requests.

- **Packages Introduced:**

- **marimo** : For creating reactive, app-like notebooks and user interface (UI) elements (text input).
- **yfinance** : The primary tool for programmatic access to financial data (Balance Sheets, Income Statements, Stock Price History) from Yahoo Finance.
- **pandas** : Briefly introduced to structure data for plotting (DataFrames) and to save data to CSV files.
- **altair** : For declarative, statistical data visualization.